

Information theory

Shannon's limit (Noisy-channel coding theorem)

The Shannon limit or Shannon capacity of a communication channel refers to the **maximum** rate of **error-free** data that can **theoretically** be transferred over a **noisy** channel.

The channel capacity C can be calculated from the physical properties of a channel; for a band-limited channel with Gaussian noise, using the **Shannon-Hartley theorem**:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

Note: S/N is the signal-to-noise ratio (SNR) or the carrier-to-noise ratio (CNR) of the communication signal to the noise and interference at the receiver (expressed as a **linear** power ratio, **NOT** as logarithmic decibels).

non-linear Shannon limit:

for optic fibers: Stimulated Brillouin scattering (SBS)

Nyquist rate (Nyquist criterion)

no noise

$f_s > f_n = 2B$, where f_n is Nyquist rate

Nyquist sampling limit: $f_s > 2 * f_h$

Nyquist sampling limit $f_s > 2f_h$

Nyquist sampling sets a limit for the **minimum sample rate** for recording a signal, this is important as sampling frequency not being correct will lead to lost data and **aliasing** effects that will result in errors of the measured signal.

where f_s is the sample frequency and f_h is the highest frequency in the wanted signal.

##

resolution: bit

voltage resolution: V

dynamic range:

$10 \log_{10} (\text{max_voltage} / \text{solution})^{**2}$

$= 20 \log_{10} (\text{max_voltage} / \text{voltage_resolution})$

$$f * g = \mathcal{F}^{-1}(\mathcal{F}(f) \cdot \mathcal{F}(g))$$

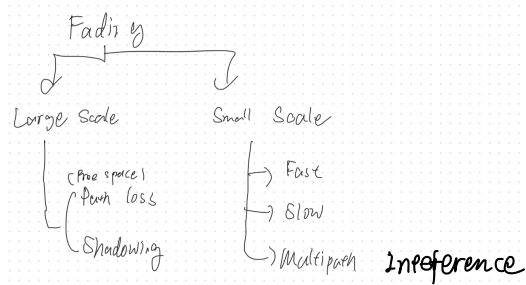
Where $\mathcal{F}(f)$ and $\mathcal{F}(g)$ are the Fourier transforms of f and g .

voltage_re

$\text{ENOB} = (S/N - 1.76) / 6.02$

$S/N = 20 \log_{10} (V_{\text{signal}} / V_{\text{noise}})$

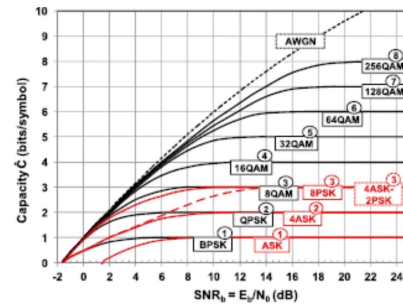
$P_{\text{p2p}} = 6 P_{\text{rms}}$



Free space path loss: A fundamental loss when a signal is transmitted over a long distance, independent of the type of transceiver.

Shadowing: loss in the signal power due to the obstructions in the path of propagation.

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Rayleigh Fading